Expressions: Find the Distribution & Like-Term Errors

Directions:

- Distribute to *every* term, and combine only *like* terms.
- Several problems show another student's work with a mistake in it — your job is to find the mistake and fix it.
- When you test two expressions with a number, remember: equivalent expressions agree for *every* value of the variable.

1. Combine like terms: $5x + 3x$

Answer: ________________

2. Expand: $6(x + 4)$

Answer: ________________

3. Combine like terms: $4x + 7 + 3x - 2$

Answer: ________________

4. Consider the two expressions $3(x+2)$ and $3x+2$.

(a) Evaluate $3(x+2)$ when $x=5$: ________

(b) Evaluate $3x+2$ when $x=5$: ________

(c) Are the two expressions equivalent? Say how you know.

5. Deshawn evaluates $4(x+3)$ when $x=2$. He writes:

$$4(x+3) = 4 \cdot 2 + 3 = 11.$$

(a) What error did Deshawn make? Answer in one sentence.

(b) Find the correct value of $4(x+3)$ when $x=2$.

Answer: ________________

6. Ava simplifies $6x-4x$ and writes:

$$6x - 4x = 2 \qquad \text{“because the } x\text{'s cancel.”}$$

(a) What error did Ava make? Answer in one sentence.

(b) Find the value of $6x-4x$ when $x=7$.

Answer: ________________

7. Expand: $-3(2x - 5)$

Answer: ______________

8. **Challenge.** Jaylen claims that $2(3x + 4) = 6x + 4$.

(a) Evaluate $2(3x + 4)$ when $x = 5$: ________

(b) Evaluate $6x + 4$ when $x = 5$: ________

(c) Explain what Jaylen forgot when he distributed, and write the correct expansion.